ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

# MECHATRONICS

## PAPER 1 · PRACTICE A · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice a.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers, sensible project decisions and neat, recognisable sketches and charts.

## QUESTION 1

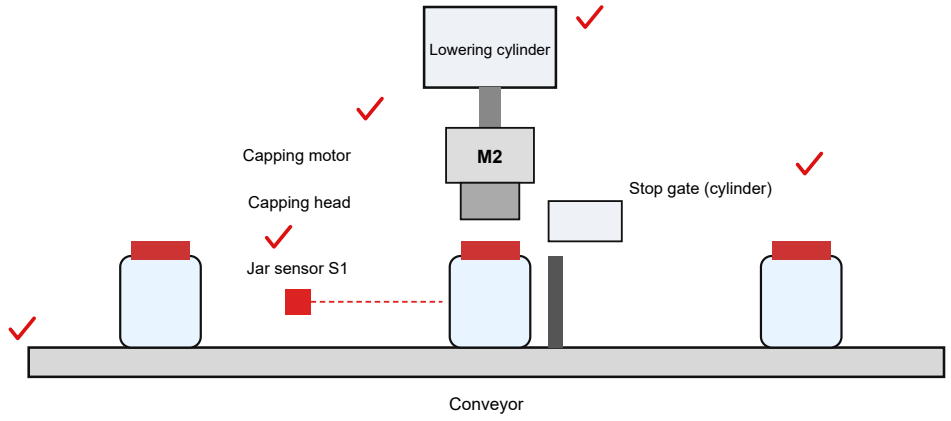
**1.1** A design brief is a short statement of the problem and what must be achieved ✓✓; a design specification is a detailed list of the exact requirements and limits the solution must meet (sizes, speeds, costs, safety) ✓✓. (4)

**1.2** Any SIX sensible specifications ✓✓✓✓✓✓, e.g. cap at least 20 jars per minute; fit jars of 60–90 mm diameter; tighten caps to a set torque without breaking the glass; run on 24 V DC controls and 6 bar air; food-safe materials that can be washed; guards and an emergency stop; cost below the budget; fit on the existing conveyor. (6)

**[10]**

## QUESTION 2

**2.1**




Neat freehand sketch ✓ showing: conveyor with jars ✓; jar sensor and stop gate ✓; lowering cylinder above the jar ✓; capping head with its motor ✓ (5)

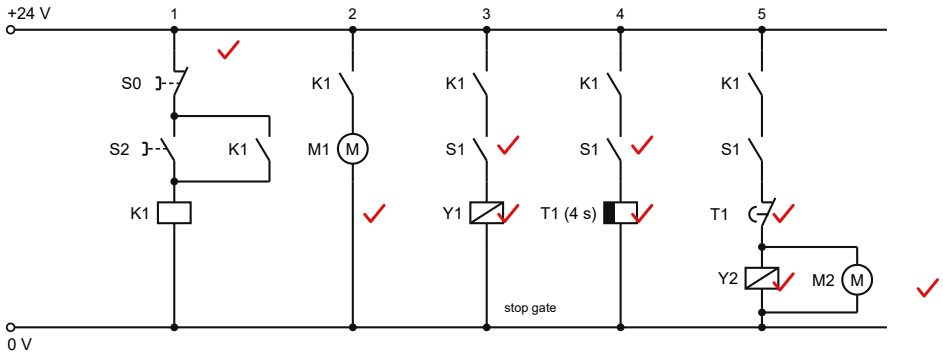
**2.2.1** $\text{Gear ratio} = (T_B \div T_A) \times (T_D \div T_C) = (60 \div 20) \times (45 \div 15)$ ✓ $= 9\ (9:1)$ ✓ (2)

**2.2.2** $N_{out} = N_{in} \div GR = 1\,400 \div 9$ ✓ $= 155{,}56\ \text{r/min}$ ✓ (2)

**2.2.3** $P_{out} = \eta \times P_{in} = 0{,}85 \times 180 = 153\ \text{W}$ ✓; $\omega = 2\pi N \div 60 = 2\pi \times 155{,}56 \div 60 = 16{,}29\ \text{rad/s}$ ✓; $T = P \div \omega$ ✓ $= 153 \div 16{,}29 = 9{,}39\ \text{N·m}$ ✓ (4)

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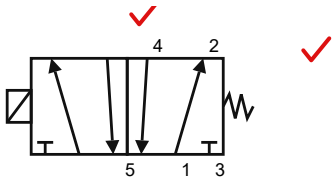
**2.3**




START / STOP latch with K1 and conveyor M1 kept ✓; S1 with a K1 contact energises Y1 (stop gate) ✓; S1 with K1 starts timer T1 ✓✓; Y2 and M2 through S1 and an NC T1 contact ✓✓ so they switch off after 4 s ✓; correct symbols and labels ✓

(8)

**2.4**




Two squares with five ports and correct flow paths ✓; solenoid on one side and spring on the other ✓

(2)

**2.5** Any TWO ✓✓: it slips when the cap is tight, so the jar or cap is not broken; it protects the motor and gears from overload; it gives every jar the same tightness. (2)

**[25]**

## QUESTION 3

**3.1**

| Item | Description | Quantity | Unit cost (R) | Total (R) |
|---|---|---|---|---|
| 1 | Pneumatic cylinder (lowering) | 1 | 1 200 | 1 200 |
| 2 | Stop-gate cylinder | 1 | 850 | 850 |
| 3 | 5/2 solenoid valves | 2 | 950 | 1 900 |
| 4 | DC gear motor and clutch | 1 | 2 400 | 2 400 |
| 5 | Jar sensor (photo-electric) | 1 | 650 | 650 |
| 6 | PLC / control relay and timer | 1 | 3 500 | 3 500 |
| 7 | Frame, brackets and fasteners | 1 | 1 800 | 1 800 |

Suitable headings ✓✓; any SIX sensible items ✓✓✓✓ (2 items per mark); totals and a grand total ✓✓

(8)

**3.2**

**Capping station – project plan**

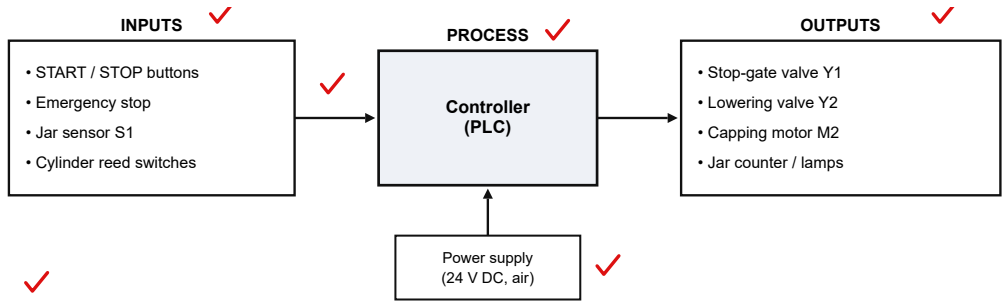
| Task | Week 1 | Week 2 | Week 3 | Week 4 |
|---|---|---|---|---|
| Design and drawings | | | | |
| Ordering parts | | | | |
| Building frame and head | | | | |
| Pneumatics | | | | |
| Wiring and programming | | | | |
| Testing and commissioning | | | | |

Week scale 1–4 along the top ✓; SIX sensible activities ✓✓ (e.g. design, ordering, building the frame and head, pneumatics, wiring and programming, testing and commissioning); bars of sensible length in a logical order ✓✓✓✓; overlapping activities ✓; neat with a title ✓

(10)

**3.3** Any THREE hazards ✓✓✓ with a matching control ✓✓✓, e.g. fingers caught under the capping head – guards and two-hand testing; broken glass – safety glasses and gloves; electric shock – isolate and lock out before wiring; compressed-air accidents – turn the air off and bleed the lines; lifting injuries – lift correctly or use help. (6)

**3.4**




Inputs ✓ (START, STOP, E-stop, jar sensor, cylinder reed switches); controller (PLC) ✓; outputs ✓ (stop-gate valve, lowering valve, capping motor, counter display / lamps); power supply ✓; arrows showing the direction of signals ✓; labels ✓

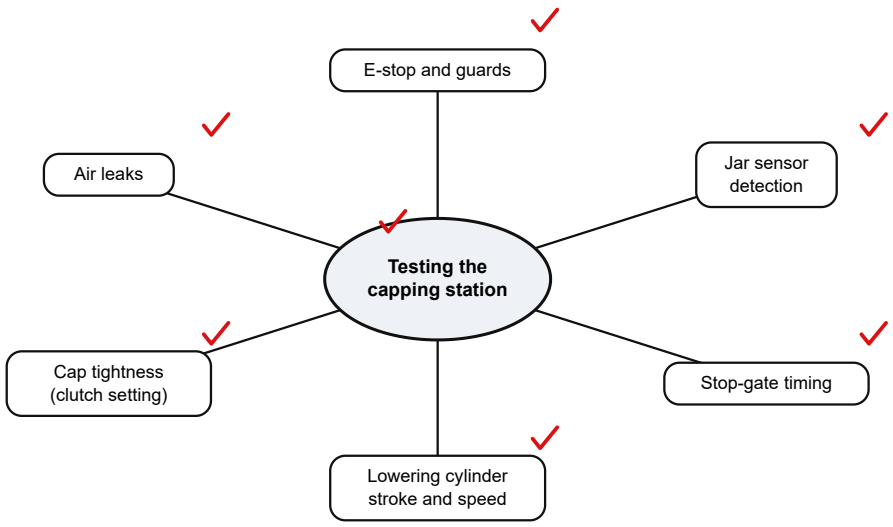
(6)

**[30]**

## QUESTION 4

**4.1** Testing checks that each part works correctly ✓; commissioning brings the whole station into service, proving that it works as specified and handing it over ✓. (2)

**4.2**




Centre topic "Testing the capping station" ✓ with any FIVE areas ✓✓✓✓✓: E-stop and guards; jar sensor detection; stop-gate timing; lowering cylinder stroke and speed; cap tightness / clutch setting; motor direction; air leaks; counter accuracy.

(6)

**4.3** Cap a sample of jars ✓, then test the opening torque with a torque tester (or by hand) ✓ and adjust the clutch / air pressure / capping time until all caps are tight without breaking ✓. (3)

**4.4** Any FOUR ✓✓✓✓: operator's manual; circuit and wiring diagrams; PLC program (printout and backup); maintenance schedule; spare parts list; test and commissioning report; warranty certificates. (4)

**[15]**

## QUESTION 5

**5.1** Make the station safe (isolate, lock out, air off where needed) ✓; check the 24 V supply to the solenoid at the valve ✓; check the wiring and connector between the PLC output and Y2 ✓; check the solenoid coil (resistance / does it click) ✓; check the air supply pressure and the valve spool / manual override ✓; check the cylinder and replace the faulty part, then test again ✓. (6)

**5.2** Breakdown: no cost or planning until a failure happens ✓✓. Preventive: fewer unexpected failures because parts are serviced at fixed times ✓✓. Predictive: parts are replaced only when monitoring shows they are wearing, so less downtime and fewer parts wasted ✓✓. (6)

**5.3.1** $\text{MTBF} = \text{operating time} \div \text{number of failures} = (720 - 12) \div 4$ ✓ $= 177$ h ✓ (accept $720 \div 4 = 180$ h and the matching answers in 5.3.3) (2)

**5.3.2** $\text{MTTR} = \text{repair time} \div \text{number of failures} = 12 \div 4$ ✓ $= 3$ h ✓ (2)

**5.3.3** $\text{Availability} = \text{MTBF} \div (\text{MTBF} + \text{MTTR}) \times 100 = 177 \div 180 \times 100$ ✓ $=$ $98{,}33\%$ ✓ (2)

**5.4** They show patterns of repeated faults ✓, which helps to plan maintenance, order spares and prove that safety checks were done ✓. (2)

**[20]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**